

Depth × Color Decoupling — Experiment Design

Independent control of depth-plane and color identity for translator and noise companion

Motivation

In DepthColorLinked (current experiment), Near=Red and Far=Green are always linked

(linkDepthColor=1). ZdCoh (ZdA) changes BOTH the depth plane and the color of the coherent

translator simultaneously; ZdNoi (ZdB) changes BOTH for the noise companion. F2 (Depth Cueing)

is therefore confounded with color change — it is impossible to determine whether the

disruption in ZdCoh is driven by depth, color, or both.

Additional confound in F1 (Dot Cueing): color reversals hit the TARGET in CUED+ZdCoh but the

DISTRACTOR in UNCUED+ZdCoh — the comparison is asymmetric in disruption, making F1 a

conservative underestimate of the true dot-cueing advantage.

Three independent factors at tStart

F1 Dot Cueing

CUED vs UNCUED — does the delayed-onset field translate?

F2 Depth Swap on translator

Does the coherent translator change depth plane at tStart? (YES → new depth plane)

F3 Color Swap on translator

Does the coherent translator change color at tStart? (YES → new color)

The noise companion is held at NO SWAP (neither depth nor color changes) throughout the primary

experiment — this isolates the translator's identity change as the sole variable. Companion-

swap conditions (ZdB-type) are listed separately on page 3 as a secondary extension.

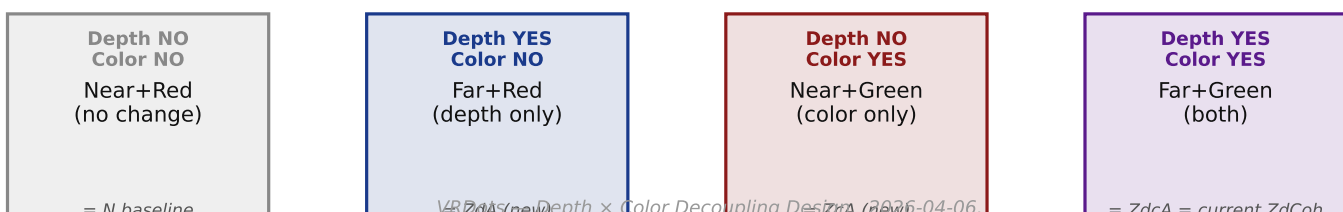
Stimulus identity: Near = Red · Far = Green (unchanged)

At trial start the translator occupies depth DFD (Near or Far, balanced) with the corresponding

color (Red or Green). At tStart it may jump to the opposite depth, switch to the opposite

color, both, or neither — as determined by F2 × F3.

Translator state at tStart (example: starts Near = Red)



DFD is balanced: half the trials start Near=Red, half start Far=Green. Swap directions are

therefore symmetric across the design.

Primary Conditions (companion = no swap throughout)

2 (Dot Cueing) × 2 (Depth Swap) × 2 (Color Swap) = 8 conditions × 2 (DFD Near/Far) = 16 cells

#	Code	F1 Dots	F2 Depth swap	F3 Color swap	Translator at tStart	Current analog	Predicted perf
1	CUED + N	CUED	✗	✗	keeps depth+color	N baseline (not run yet)	High ~+35pp
2	CUED + ZdA	CUED	✓	✗	→ opp. depth same color	New	? depth only disruptive?
3	CUED + ZcA	CUED	✗	✓	same depth → opp. color	New	? color only disruptive?
4	CUED + ZdC A	CUED	✓	✓	→ opp. depth → opp. color	= current ZdCoh (ZdA+color)	Reduced ~+18pp
5	UNCUED + N	UNCUED	✗	✗	keeps depth+color	N baseline (not run yet)	Low ~+10pp
6	UNCUED + ZdA	UNCUED	✓	✗	→ opp. depth same color	New	?
7	UNCUED + ZcA	UNCUED	✗	✓	same depth → opp. color	New	?
8	UNCUED + ZdC A	UNCUED	✓	✓	→ opp. depth → opp. color	= current ZdCoh (UNCUED)	Low ~+9pp

Predicted patterns by hypothesis

Depth-driven:

Cond 2 ≈ Cond 4 (ZdA ≈ ZdC A, both disrupted); Cond 3 ≈ Cond 1 (ZcA ≈ N, both spared).
F2 survives when depth changes; F3 null when color changes alone.

Color-driven:

Cond 3 ≈ Cond 4 (ZcA ≈ ZdC A, both disrupted); Cond 2 ≈ Cond 1 (ZdA ≈ N, spared).
F3 survives; F2 null.

Additive:

Cond 4 < Cond 2 < Cond 1 and Cond 4 < Cond 3 < Cond 1.
Both F2 and F3 significant; ZdC A worst.

Interaction:

Cond 2 and Cond 3 each produce partial disruption; Cond 4 disproportionately worse than either alone (depth × color interaction).

Trial counts

VRDots — Depth × Color Decoupling Design. 2026-04-06.

8 conditions × 2 DFD (Near/Far) = 16 cells. At 32 trials/cell → 512 trials/session (same as DepthColorLinked). Two sessions → n=64/cell, sufficient to detect ≥15pp differences at α=0.05 with ~80% power. F1 is now clean: both CUED and UNCUED sides include N (no swap) and matched swap conditions, eliminating the target-vs-distractor asymmetry present in the current design.

Secondary Conditions (companion swap, translator = no swap)

Mirrors page 2 but varies the noise companion instead of the translator

#	Code	F1 Dots	F2 Depth swap	F3 Color swap	Translator at tStart	Current analog	Predicted perf
9	CUED + N	CUED	X	X	companion: no change	= primary #1	same cell
10	CUED + ZdB	CUED	✓	X	companion: → opp. depth	New	? companion depth only
11	CUED + ZcB	CUED	X	✓	companion: → opp. color	New	? companion color only
12	CUED + ZdcB	CUED	✓	✓	companion: → depth+color	= current ZdNoi	High ~+35pp
13	UNCUED + N	UNCUED	X	X	companion: no change	= primary #5	same cell
14	UNCUED + ZdB	UNCUED	✓	X	companion: → opp. depth	New	?
15	UNCUED + ZcB	UNCUED	X	✓	companion: → opp. color	New	?
16	UNCUED + ZdcB	UNCUED	✓	✓	companion: → depth+color	= current ZdNoi UNCUEdB	High ~+11pp

Combined primary + secondary: new SwapFlags needed

- Color50A (ZcA)

Color swap on coherent translator only (no depth change)
- Color50B (ZcB)

Color swap on noise companion only (no depth change)
- Depth50A (ZdA)

Depth swap on coherent translator — ALREADY IMPLEMENTED
- Depth50B (ZdB)

Depth swap on noise companion — ALREADY IMPLEMENTED

ZdcA = Depth50A | Color50A (translator changes both). ZdcB = Depth50B | Color50B (companion changes both) = current ZdNoi. Bit-combining existing and new flags gives all 16 conditions without new enum values beyond Color50A and Color50B.

Recommended run order

Phase 1 (primary, 8 conditions): Run 2 sessions (~512 trials each) to answer depth vs color on the TRANSLATOR. This directly resolves the ZdCoh confound. Include N (no swap) baseline — currently absent from DepthColorLinked.

Phase 2 (secondary, 6 new companion conditions): Add ZdB, ZcB to test whether companion identity also matters. Can be combined with Phase 1 replication for efficiency (16 conditions × 2 DFD × 16 trials = 512 trials/session).

Both phases keep Near=Red / Far=Green and depth sep = 0.05 m to maintain comparability with DepthColorLinked sessions.